COMPUTACIÓN DE ALTAS PRESTACIONES (CAP)

Tema 0: Software & Hardware

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Objetivos

- Aprender los conceptos fundamentales en la Computación de Altas Prestaciones (**HPC**)
  - Conocimientos sobre arquitectura de procesadores
  - Conocimientos en sistemas **homogéneos** y **heterogéneos**
  - Desarrollo de códigos paralelos y optimizados
    - **OpenMP**
    - **CUDA**
    - Instrucciones Vectoriales (**SIMD**)

Requisitos

- Conocimientos básicos:
  - Lenguajes de programación
  - Álgebra lineal

~~python~~ ~~Java~~ C

- Material:
  - Jetson Nano 2GB (hacer parejas y una Jetson Nano por cada una de ellas)
  - **Memoria USB (Backups + entregar los códigos desarrollados)**

Bibliografía

OPENMP
https://www.openmp.org/resources/refguides/

CUDA
https://docs.nvidia.com/cuda/cuda-c-programming-guide/index.html

ARM DEVELOPER
https://developer.arm.com/

MATERIAL COMPLEMENTARIO

- **David B. Kirk and Wen-mei W. Hwu. 2016. Programming Massively Parallel Processors, Third Edition: A Hands-on Approach (3rd. ed.). Morgan Kaufmann Publishers Inc., San Francisco, CA, USA**

- **David R. Kaeli, Perhaad Mistry, Dana Schaa, and Dong Ping Zhang. 2015. Heterogeneous Computing with OpenCL 2.0 (1st. ed.). Morgan Kaufmann Publishers Inc., San Francisco, CA, USA**

Empezamos…

1. Parejas para la evaluación
2. Asignación de Jetson Nano por parejas
3. GEMM Driver:
   https://github.com/martineh/driverCAP.git

Guía GEMM Driver

Línea de ejecución

```
nano-0@nano3-desktop:~/Desktop/CAP/gemm_driver$ ./run_driver.sh
make: Warning: File 'Makefile' has modification time 140747 s in the future
rm build/*
make: warning:  Clock skew detected.  Your build may be incomplete.
make: Warning: File 'Makefile' has modification time 140747 s in the future
gcc -O3 -fopenmp -Wall -fPIC -std=c99 -march=armv8-a  gemm_driver.c my_gemm.c lib/libgemm.a -o build/gemm_driver.x -I./include -lm
make: warning:  Clock skew detected.  Your build may be incomplete.

[*] Level Selected     : VERY HARD (100%)!!
[*] Data Tesing        : FLOATS
[*] Convolutions GEMM  : ResNet50v15_imagenet
```

D R I V E R   F O R   M A T R I X   M U L T I P L I C A T I O N

MATRIX DIMENSIONS			YOUR IMPLEMENTATION		GOAL IMPLEMENTATION		TEST
m	n	k	GFlops	Test Error	GFlops	Test Error	WIN
12544	64	147	0.42	0.00e+00	8.63	5.61e-08	NO
3136	64	64	0.48	0.00e+00	9.90	5.26e-08	NO
3136	64	576	0.29	0.00e+00	10.22	3.73e-07	NO
3136	256	64	0.26	0.00e+00	14.65	5.32e-08	NO
3136	64	256	0.30	0.00e+00	8.74	5.30e-08	NO
3136	128	256	0.28	0.00e+00	13.85	5.36e-08	NO
784	128	1152	0.26	0.00e+00	12.66	5.19e-07	NO
784	512	128	0.27	0.00e+00	13.80	5.34e-08	NO
784	512	256	0.28	0.00e+00	11.09	5.32e-08	NO
78[illegible]	[illegible]	512	0.28	0.00e+00	11.21	3.28e-07	NO
784	256	512	0.27	0.00e+00	10.75	3.25e-07	NO
196	256	2304	0.14	0.00e+00	10.35	7.24e-07	NO
196	1024	256	0.24	0.00e+00	13.37	5.35e-08	NO
196	1024	512	0.12	0.00e+00	12.87	3.25e-07	NO
196	256	1024	0.28	0.00e+00	11.20	4.45e-07	NO
196	512	1024	0.15	0.00e+00	11.47	4.46e-07	NO
49	512	4608	0.08	0.00e+00	6.16	1.02e-06	NO
49	2048	512	0.09	0.00e+00	6.01	3.26e-07	NO
49	2048	1024	0.04	0.00e+00	5.30	4.45e-07	NO
49	512	2048	0.08	0.00e+00	5.53	6.11e-07	NO
N O T P A S S!			0.23		10.39		

Capas convolucionales **ResNet50v15**

Has superado o no el objetivo

Rendimiento de tu GEMM

Objetivo (Según nivel dificultad)

¿En qué consiste?

Guía GEMM Driver

```
[*] Level Selected      : VERY EASY (20%)
[*] Data Tesing         : FLOATS
[*] Convolutions GEMM   : ResNet50v15_imagenet
```

D R I V E R   F O R   M A T R I X   M U L T I P L I C A T I O N

MATRIX DIMENSIONS			YOUR IMPLEMENTATION		GOAL IMPLEMENTATION		TEST
m	n	k	GFlops	Test Error	GFlops	Test Error	WIN
12544	64	147	0.32	0.00e+00	0.07	0.00e+00	YES
3136	64	64	0.52	0.00e+00	0.14	0.00e+00	YES
3136	64	576	0.29	0.00e+00	0.10	0.00e+00	YES
3136	256	64	0.26	0.00e+00	0.08	0.00e+00	YES
3136	64	256	0.27	0.00e+00	0.08	0.00e+00	YES
3136	128	256	0.27	0.00e+00	0.08	0.00e+00	YES
784	128	1152	0.26	0.00e+00	0.13	0.00e+00	YES
784	512	128	0.28	0.00e+00	0.15	0.00e+00	YES
784	512	256	0.26	0.00e+00	0.15	0.00e+00	YES
784	128	512	0.28	0.00e+00	0.13	0.00e+00	YES
784	256	512	0.27	0.00e+00	[illegible]	[illegible]	YES
196	256	2304	0.14	0.00e+00	0.15	0.00e+00	NO
196	1024	256	0.26	0.00e+00	0.16	0.00e+00	YES
196	1024	512	0.12	0.00e+00	0.15	0.00e+00	NO
196	256	1024	0.29	0.00e+00	0.15	0.00e+00	YES
196	512	1024	0.15	0.00e+00	0.14	0.00e+00	YES
49	512	4608	0.08	0.00e+00	0.14	0.00e+00	NO
49	2048	512	0.07	0.00e+00	0.13	0.00e+00	NO
49	2048	1024	0.04	0.00e+00	0.12	0.00e+00	NO
49	512	2048	0.09	0.00e+00	0.14	0.00e+00	NO
P A S S!			0.23		0.13		

1.Validación Resultado

2.Superar Gflops Referencia

3.Superar Gflops rendimiento global

¿Cómo superar un nivel?

Error en la GEMM ¿Qué hago?

```
nano-0@nano3-desktop:~/Desktop/CAP/gemm_driver$ ./run_driver.sh
rm build/*
gcc -O3 -fopenmp -Wall -fPIC -std=c99 -march=armv8-a  gemm_driver.c my_gemm.c lib/libgemm.a -o build/gemm_driver.x -I./include -lm

[*] Level Selected     : VERY EASY (20%)
[*] Data Tesing        : FLOATS
[*] Convolutions GEMM  : ResNet50v15_imagenet

=================================================================================================
|               D R I V E R   F O R   M A T R I X   M U L T I P L I C A T I O N               |
=================================================================================================
|       MATRIX DIMENSIONS          |     YOUR IMPLEMENTATION      |     GOAL IMPLEMENTATION      | TEST |
+----------------------------------+------------------------------+------------------------------|      |
|      m         n         k       |   GFlops       Test Error    |   GFlops       Test Error    |  WIN |
+----------------------------------+------------------------------+------------------------------+------+
|   12544        64       147      |    0.24        1.00e+00
nano-0@nano3-desktop:~/Desktop/CAP/gemm_driver$
```

Guía GEMM Driver

```
#+----------------------------------------------------+#
#|            R U N    C O N F I G U R E              |#
#+----------------------------------------------------+#
#+----------------------------------------------------+#
#---------------------------------
#| VALUE | DATASET               |
#-------+-------------------------
#|   0   | TESTING               |
#-------+-------------------------
#|   1   | RESNET50V15_IMAGENET |
#---------------------------------
DATASET=1
#---------------------------------


#-------------------------
#| VALUE | PRINT MATRIX |
#-------+-----------------
#|   0   | OFF           |
#-------+-----------------
#|   1   | ON            |
#-------------------------
VISUAL=0
#-------------------------


#---------------------------------------
#| VALUE | LEVEL DIFFICULTY    | SCORE |
#-------+---------------------+---------
#|   0   | VERY EASY           | 20%   |
#-------+---------------------+---------
#|   1   | EASY                | 40%   |
#-------+---------------------+---------
#|   2   | MEDIUM              | 60%   |
#-------+---------------------+---------
#|   3   | HARD                | 80%   |
#-------+---------------------+---------
#|   4   | VERY HARD           | 100%  |
#---------------------------------------
LEVEL=4
#---------------------------------------


#+----------------------------------------------------+#
#+----------------------------------------------------+#

make clean; make && ./build/gemm_driver.x $VISUAL $LEVEL $DATASET
```

Activar Modo visual y Dataset para Testing

Guía GEMM Driver

```
=====================================================================================
|                D R I V E R   F O R   M A T R I X   M U L T I P L I C A T I O N    |
=====================================================================================
|     MATRIX DIMENSIONS          |    YOUR IMPLEMENTATION     |    GOAL IMPLEMENTATION     | TEST |
+--------------------------------+----------------------------+----------------------------|      |
|    m        n        k         |  GFlops       Test Error   |  GFlops       Test Error   | WIN  |
+--------------------------------+----------------------------+----------------------------+------+
```

```
+----------+
| TENSOR A |
+----------+
0.840 0.394 0.783 0.798 0.912
0.198 0.335 0.768 0.278 0.554
0.477 0.629 0.365 0.513 0.952
0.916 0.636 0.717 0.142 0.607
0.016 0.243 0.137 0.804 0.157
```

Tensor/Matriz **A**

```
+----------+
| TENSOR B |
+----------+
0.401 0.130 0.109 0.999 0.218
0.513 0.839 0.613 0.296 0.638
0.524 0.494 0.973 0.293 0.771
0.527 0.770 0.400 0.892 0.283
0.352 0.808 0.919 0.070 0.949
```

Tensor/Matriz **B**

```
+----------+
| TENSOR C |
+----------+
0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000 0.000
```

Tensor/Matriz **C** (tu resultado)

```
+----------+
| TENSOR Ct|
+----------+
1.692 2.178 2.252 1.961 2.131
0.995 1.347 1.594 0.808 1.454
1.311 1.934 1.873 1.294 1.836
1.358 1.606 1.801 1.482 1.775
0.682 1.019 0.750 0.856 0.641
```

Tensor/Matriz **Ct** (referencia, valores correctos)

```
|     5        5        5     |    0.00      1.00e+00
nano-0@nano3-desktop:~/Desktop/CAP/gemm_driver$
```

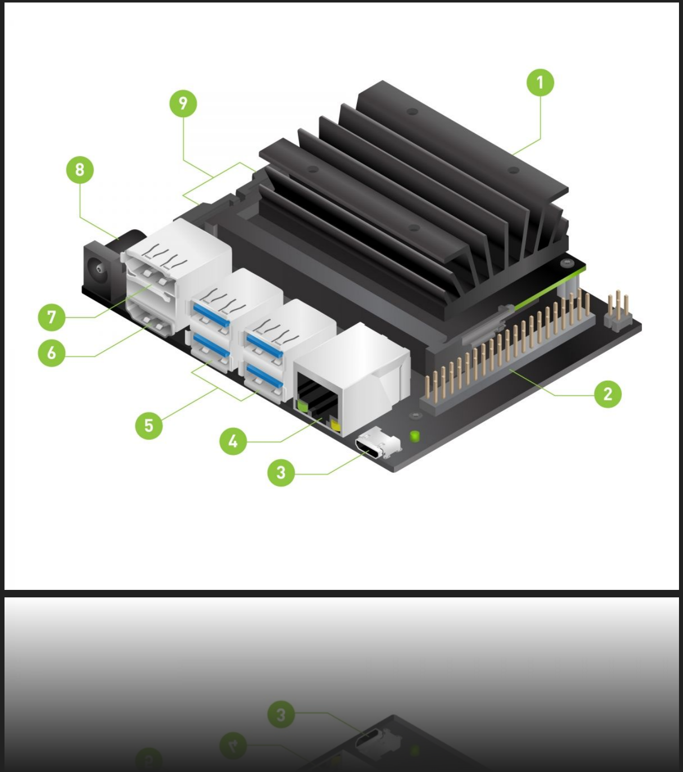
Visualizar valores Tensores y resultados

Nvidia Jetson Nano (Conociendo Nuestro Hardware...)

ESPECIFICACIONES

Arquitectura	**ARM Cortex A-57**
Procesadores	**4 x 1,43 GHz**
RAM/GPU Memory	**2 GB LPDDR4**
GPU	**128 núcleos**
Instrucciones Vectoriales	**ARM Neon v8**

Nvidia Jetson Nano (Conociendo Nuestro Hardware...)



1	microSD card slot for main storage
2	40-pin expansion header
3	Micro-USB port for 5V power input, or for Device Mode
4	Gigabit Ethernet port
5	USB 3.0 ports (x4)
6	HDMI output port
7	DisplayPort connector
8	DC Barrel jack for 5V power input
9	MIPI CSI-2 camera connectors

¿Preguntas?